

Day-Ahead Travel Time Forecasting on Freeway 5 Southbound with an Interpretable Linear Model

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Code and data: <https://github.com/leeshuncher/DataMiningHW1>

Abstract

Drivers from Taipei to Yilan decide the evening before when to leave, yet the Hsuehshan Tunnel bottleneck is congested only on certain days and hours. We forecast hourly southbound travel time on Freeway 5 one day ahead from calendar and holiday-structure features alone, with a median linear regression of 109 terms whose coefficients read as minutes of delay. Trained on 2023–2025 with every choice made by rolling cross-validation, and scored on a pre-registered blind test (January–August 2026), it reaches MAE 1.49 min, 26% below last week’s same hour, and ties the strongest baseline, gradient boosting (L1), overall. On long weekends it cuts error by 39% (95% CI 26–48%), beats that baseline on first-day afternoons by 32% and in congested hours by 9%, and its recommended departure hour is within 0.45 min of the best, well inside our 1.0 min threshold. A 2025 development test gives the same picture.

1 Introduction

Freeway 5 is the main fast route between Taipei and Yilan; its southbound direction funnels into the Hsuehshan Tunnel. Demand is mostly leisure travel, so queues follow the calendar: Saturday afternoons, the first day of a long weekend, the evening before a holiday.

User and decision. A driver (or trip app) chooses the evening before which departure hour to take tomorrow. Predictions are therefore made at 18:00 of the previous day from information known then.

Research question. Can a calendar-based linear regression forecast next-day southbound travel time well enough to choose a departure hour, and where does it fail?

Contributions. (1) A decision metric in minutes: the time lost by leaving at the recommended hour instead of the truly fastest hour in a 4-hour window, threshold 1.0 min. (2) A compact 109-term linear model whose coefficients are minutes of delay;

on a pre-registered blind 2026 test it ties gradient boosting overall, beats it where congestion occurs, and cuts last week’s long-weekend error by 39%. (3) An analysis of what the model learned: long-weekend congestion is an afternoon effect (first-day mornings are *faster* than Saturday mornings), eve evenings are 12–29% slower, and the remaining large errors are incidents that no calendar can foresee.

2 Related Work

Table 1 compares four verified sources. Rice and van Zwet (2004) rely on the *current* travel time, which does not exist the evening before; Fan et al. (2018) use the same TDCS data with random forests on hour and weekday but no holiday features; Ashouri et al. (2023) is closest in spirit, a calendar-driven OLS for Taiwanese holidays, but forecasts vehicle counts. We differ in target (travel time through the Freeway 5 bottleneck), horizon (one day, calendar only) and evaluation (a departure decision).

3 Task Definition and Data

Task. Table 2 defines the task: one forecast per departure hour, made the evening before from facts that are already fixed then (the official holiday calendar is published months ahead).

Data. TDCS M04A (Freeway Bureau, MOTC, 2025) (median segment travel times per 5 min, public) for 2023-01-01 to 2025-12-31 (1,096 days, 26,304 hourly windows, 37 missing). We keep Freeway 5 southbound small cars (type 31), the vehicle class of our users. We drop invalid rows (seconds > 0, count ≥ 1 required) because they encode missing detections, not fast trips. We sum the four segment medians within the same 5-min slot to obtain a trip time (missing if any segment is invalid, so no partial trip is counted), and average to hours with at least 6 of 12 valid slots so that one or two

Work	Data	Target	Features	Model	Reported error
Rice and van Zwet (2004)	I-10 LA, 48 mi, 34 weekdays	travel time, 0–60 min ahead	current travel time, time of day	time-varying linear	RMSE <10 min
Fan et al. (2018)	TDCS, 2015/2016	travel time	hour, weekday; real-time update	random forest (Hadoop)	MAPE $\approx$ 5%
Bansal (2020)	Freeway 1, 36.1 km	travel time	43 predictors $\rightarrow$ 6	BBO-SVR	MAPE 6.44%
Ashouri et al. (2023)	Taiwan holidays 2019–21	hourly vehicle counts	Fourier terms, hierarchy	reconciled OLS	RMSE per level, OLS $\approx$ ARIMA
Ours	Freeway 5 SB, 2023–25	travel time, 1 day ahead	calendar only	median linear, 109 terms	MAE 1.42 min, MAPE 3.6%

Table 1: Related work. Errors are on different data and horizons, so they give context, not a ranking.

Item	Definition
Target	Mean small-car travel time (min), Nangang System (km 0) to 05F0439S (km 43.9), southbound, through the Hsuehshan Tunnel; observed range 27.6–87.8 (1st–99th pct. 28.4–57.9)
One row	One departure hour (24 per day)
Inputs	Hour, weekday, holiday type and position in the holiday block, make-up workday
Known at	18:00 the day before; no same-day traffic, no weather
Data	TDCS M04A 2023–2025, 26,304 hourly rows

Table 2: Task card.

Split	Days	Rows	Mean	SD	>40 min
Train 2023	337	8,053	32.3	6.6	8.8%
Train 2024	366	8,679	31.5	4.8	5.0%
Test 2025	365	8,715	31.7	5.2	5.7%

Table 3: Target summary (minutes).

slots cannot define an hour. Nothing is imputed: only 0.14% of hours are missing, and imputing them would invent exactly the rare jams we want to measure. The sum ignores that a 30-min trip crosses slots, and the endpoint is 2.6 km past the Toucheng exit. Holidays and make-up workdays come from TaiwanCalendar (ruyut, 2025); a long weekend is ≥ 3 consecutive non-working days and the eve is the working day before. All methods are scored on identical rows (target and four weeks of history present).

4 Exploratory Feature Analysis

The shape of the day depends on the day type (Fig. 1): weekdays peak at 11:00 (38.1 min), Saturdays at 13:00 (48.5), the first day of a long weekend at 15:00 (49.0), and the eve rises late (46.4 at 23:00). The share of hours above 40 min is 3.7% on weekdays, 25.3% on Saturdays, 22.9% on first days. Saturdays drift: 30.3% in 2023 but 21.4% in 2024.

Group	Terms	Hyp.
Hour of day	23 hourly effects (Fig. 1a)	–
Day type $\times$ block	9 day types (Sat, Sun, eve Fri/other, LW first/mid/last, single holiday, make-up day), each main effect + 8 two-hour blocks 06–22h	H1–3
Friday evening	4 terms (Fri $\times$ 14–22h blocks)	H3
Long holiday	1 term: first day, 12–18h, block ≥ 4 days	H4

Table 4: Feature groups; $23+81+4+1 = 109$.

Association. Last week and the 4-week mean have the highest Pearson/Spearman/MI with the target (r 0.60–0.67, MI 0.74–0.80); hour of day has the highest MI among calendar variables (0.73). Holiday indicators look weak ($|r| \leq 0.14$) only because they cover few rows (10 first days and 10 eves in 703 days) yet their conditional effect is large; a correlation ranking would discard exactly the features that matter on long weekends (tested in §7). Redundancy: last week vs. 4-week mean $r=0.81$, Saturday vs. Saturday $\times$ daytime 0.62, so we check VIFs in the final model (max 3.1).

Hypotheses. H1: the first morning of a long weekend is much slower than an ordinary Saturday morning. H2: the last day barely affects southbound traffic. H3: the evening before a holiday is slower than an ordinary evening of the same weekday. H4: the first-day afternoon peak is lower for holidays of ≥ 4 days.

5 Method

Features. All features are 0/1 indicators of facts known the evening before, so every coefficient is minutes of delay (Table 4); after dropping empty columns the model has 109 terms. Last week, the 4-week mean and same-holiday-last-time have the strongest marginal association; adding them improved CV score by only 0.1%, below our 2%

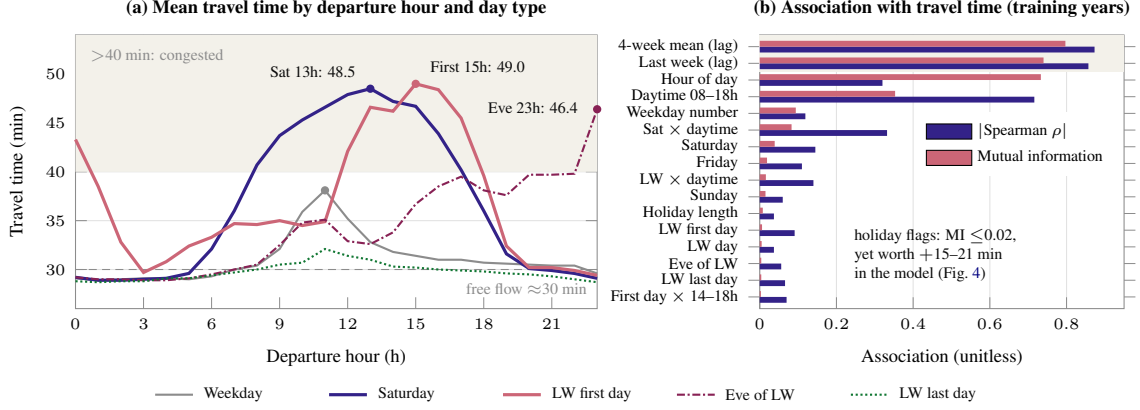


Figure 1: Exploratory analysis on 2023–2024 only (LW: long weekend; eve: working day before it). (a) The day type changes *when* the day is slow, not only *how* slow: weekdays peak at 11h, Saturdays at 13h, first days at 15h, eves late at night, while the last day stays near weekdays (H1–H3). (b) Shaded rows are lag features, strongest but not adopted (§5). Holiday indicators have almost no marginal association because they cover few rows, so a correlation ranking would discard them; this motivates the day-type×block design and Set B in §7.

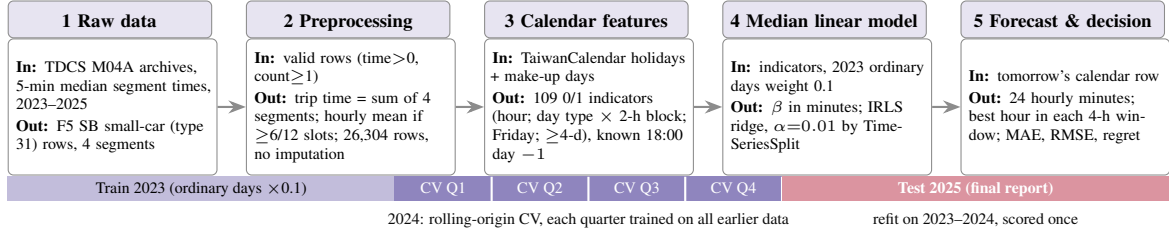


Figure 2: Pipeline (top: each stage's input and output) and data split (bottom). Every learned quantity, including the selection of features, penalty and loss, uses 2023–2024 only.

adoption rule, so they are excluded. Weather is unavailable.

Model. Median (L1) linear regression fitted by iteratively reweighted ridge (8 iterations). Two properties of the data motivate it: the target is right-skewed (free flow 30 min, rare jams up to 88 min, Table 3), so a squared loss would let a few incident hours pull every coefficient, while the median is robust to them; and the indicators are only weakly collinear ($VIF \leq 3.1$), so strong shrinkage is unnecessary. Cross-validation confirmed L1 over a log-target least-squares fit (selection score 1.97 vs. 2.25). The ridge penalty α was selected from $\{0.01, 0.1, 1, 3, 10\}$ by 4-fold TimeSeriesSplit inside the training years; the tuning curve (CV MAE 1.661, **1.659**, 1.670, 1.741, 1.806, 1.903 for $\alpha=0.001, \dots, 10$; 2.213 at 100) shows that the chosen $\alpha=0.01$ is the minimum of an extended grid and that larger penalties over-shrink the rare holiday terms, so the model is essentially unpenalised. 2023 weekdays, Saturdays and Sundays get weight 0.1 because 2023 Saturdays were much more congested; holiday days keep weight 1. Selection score

= mean of ordinary-day and holiday-day MAE; a compact design is eligible if within 5% of the best large one-hot model (2.001), and among eligible designs we take the lowest cross-validated departure loss. Blocks only: score 1.928, loss 0.715; blocks + hourly effects (chosen): 1.906, 0.304.

6 Experimental Setup

Protocol. Design selection: train 2023–2024, development test 2025. Final test: the frozen design refitted on 2023–2025 and scored once on January–August 2026, a blind test whose plan (docs/BLIND_TEST_2026.md) was committed before any 2026 prediction was made. Rolling-origin CV on the four 2024 quarters, each trained on all earlier data, never on its future (Fig. 2, bottom), with the same folds, rows and weights for every method; we report mean $\pm$ SD. Quarters are the natural fold size: each is long enough (about 2,100 hours) to contain several weekly cycles, and four folds validate every season once; all 2024 long weekends fall in Q1–Q2, so holiday CV numbers rest on those two folds. Significance: paired t -test

over folds (low power) and a day-block bootstrap (2,000 resamples of whole days) on the test year; coefficient CIs use 300 day-resampled refits.

Metrics. MAE is the primary metric because a driver’s cost is the minutes lost, linearly; RMSE is reported because rare long jams are what drivers fear most; R^2 shows variance explained beyond the mean. The decision regret measures the decision itself: for each 4-hour window (06–10, 10–14, 14–18, 18–22h) recommend the hour with the lowest predicted time; regret = actual time there minus the window minimum. **Threshold:** mean regret ≤ 1.0 min on long-weekend days, about 3% of the 30-min free-flow trip; we assume a driver cannot act on smaller differences, so below it following the forecast is practically as good as knowing the future.

Baselines. Trivial training mean; last week same hour; same holiday last time (else last week); seasonal mean hour $\times$ weekday; simple OLS on last week; gradient boosting (L1 and log targets, LightGBM, same weights); a large one-hot ridge (over 1,000 hour $\times$ day-type terms).

Protocol note. 2025 was consulted several times during development (topic check, eve variants, slot length, a first compact design); all choices were then re-made by CV on 2023–2024, but 2025 is a development test, hence the blind 2026 test.

7 Results

Main comparison. Table 5: on the 2025 development test our model is 25% (0.48 min, 95% CI 16–34%) below last week’s same hour, 50% below the training mean, and better in MAE than every simple baseline (trivial, persistence, seasonal mean, single-feature OLS), both in CV and on 2025. It has the best cross-validated holiday MAE of all methods (2.46 vs. 2.74 for GBM). Against the strongest baseline, GBM (L1), the CV difference is 0.01 min (paired t -test $p=0.72$, $n=4$); on 2025 the two differ by 0.04 min (CI 0.01–0.08), 4% of the 1.0 min threshold, so in practice they are tied, and ours exposes every effect as a readable coefficient. The large one-hot ridge with over 1,000 terms is within 1.2% in MAE (CI -1.8 to 4.0%), so a design ten times smaller gives up no accuracy.

Blind test 2026. Table 6 confirms the development results on data that played no role in any decision. Ours is 26% below last week (CI 15–36%) and 23% below the seasonal mean (CI 18–28%); against GBM (L1) the gap is 0.04 min (CI -0.02 to 0.09),

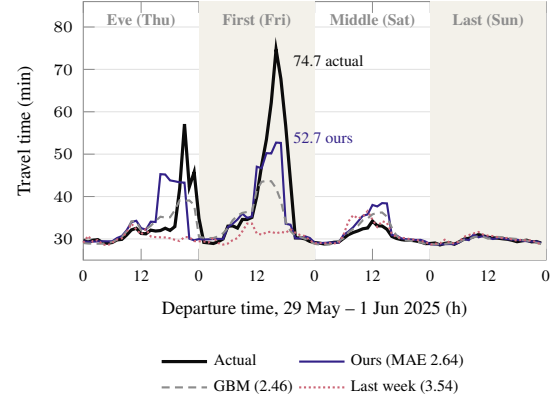


Figure 3: Hourly forecasts over the 2025 Dragon Boat long weekend and its eve (MAE over the 96 hours in the legend). The model predicts the first-day 12–18h jam from the calendar alone; last week’s same hour cannot.

a tie. Where congestion occurs we are ahead of GBM (L1): long-weekend daytime 18% (CI 2–31%), first-day afternoons 32% (CI 7–52%), hours above 40 min 9% (CI 5–13%); its edge comes from free-flow ordinary days (0.06 min, CI 0.01–0.11), the same pattern as in 2025. Long-weekend regret is 0.45 min (90% of windows within 1 min), inside the threshold. The 2023–2024 fit gives MAE 1.51 and regret 0.55: the refit does not drive the result.

Where the gain comes from. The gain is largest exactly where the user’s decision is hardest: on long-weekend days MAE falls from 3.40 to 1.69 min (50%, CI 39–59%), and on ordinary days from 1.74 to 1.35 (22%, CI 12–32%). Figure 3 shows a typical long weekend (Dragon Boat 2025, ranked 4th of 8 by our error). Last week’s value misses the holiday entirely; our model places the first-day afternoon peak at the right hours and comes closest to its height (52.7 vs. 74.7 min actual; GBM 42.0, last week 31.3).

Decision metric. On 2025, recommending the hour by our forecast saves 1.09 min per window over leaving at the window start (CI 0.95–1.24) and 1.54 on long weekends (CI 0.98–2.09); long-weekend regret is 0.82 min (last week 1.81), inside the threshold. GBM ranked first-day hours better in 2025 (0.60 vs. 2.52, 8 days), but not in 2026 (1.09 for both).

Feature ablation. Table 7 varies only the feature set (same solver, weights, folds). Set A is hour, weekday and hour $\times$ weekday; B adds the 30 holiday/structure columns most correlated with the target (selected inside each training fold; $k=30$ is the middle of the tested values; $k=15$ and 60 give CV 1.77 and 1.45, so B improves with k but stays

Model	CV MAE	CV RMSE	CV hol. MAE	Test MAE	Test RMSE	Test R^2	Test hol. MAE
Trivial: training mean	3.00±0.48	4.80±0.84	3.76±0.18	2.82	5.21	0.00	3.52
Last week, same hour	1.73±0.37	4.60±0.84	3.01±0.47	1.90	4.99	0.08	3.23
Same holiday last time	1.75±0.32	4.66±0.78	3.21±1.48	1.80	4.77	0.16	2.34
Seasonal mean: hour×weekday	1.86±0.44	3.68±0.54	2.94±0.79	1.74	4.09	0.38	3.24
Simple: OLS on last week	2.22±0.45	4.08±0.64	3.24±0.08	2.10	4.39	0.29	3.19
Ours: compact median linear	1.41±0.23	3.79±0.66	2.46±0.19	1.42	3.93	0.43	2.00
Strong: GBM (L1)	1.40±0.25	3.77±0.72	2.74±0.03	1.38	3.93	0.43	1.83
Strong: GBM (log)	1.55±0.31	3.78±0.76	3.79±0.65	1.50	3.78	0.47	2.17
Flexible linear: large one-hot ridge	1.41±0.16	3.63±0.50	2.46±0.41	1.44	3.89	0.44	2.12

Table 5: Cross-validated (mean±SD over four folds) and 2025 development-test results, minutes; best in bold. Holiday = long-weekend days and eves.

Model	LW		First	>40	LW
	MAE	days	10–18h	min	regret
Last week	2.02	3.23	18.30	13.85	1.70
Seasonal mean	1.94	3.76	16.33	13.28	1.86
Same holiday last yr	1.89	2.14	10.56	12.01	0.71
GBM (L1)	1.46	2.25	12.47	14.46	0.47
GBM (log)	1.63	2.13	8.92	12.01	0.46
Large one-hot ridge	1.52	2.14	10.28	12.57	0.46
Ours	1.49	1.97	8.52	13.15	0.45

Table 6: Pre-registered blind test, January–August 2026 (243 days; 22 long-weekend days, 5 first days). All methods refitted on 2023–2025 with the frozen design; MAE and regret in minutes.

above C at every k); C is the full model. C is 15% better than A (CI 9–21%) and 8% better than B (CI 6–10%) on the test year, with lower CV MAE as well (1.41 vs. 1.56 and 1.55). A correlation ranking finds holiday columns but not *when* in the day their delay appears. The long-weekend terms are the most valuable group for the user: on first-day afternoons they cut error from 15.1 to 6.3 min (58%), and on long-weekend daytime hours from 5.5 to 3.0 min. Saturday and Friday terms carry ordinary days (9.0% and 2.7% of overall MAE). The ≥ 4 -day and eve terms change overall MAE by less than 1% (CIs include 0); we keep them because cross-validation selected them and because they encode H3 and H4, which §8 tests directly. *Cost*: the engineered features need no extra data (the calendar is public months ahead), keep the model at 109 readable terms instead of over 1,000, and require no real-time feed.

8 Analysis

Coefficients. Figure 4 shows every term with its interval; bootstrap p -values and VIFs for all 109 terms are in `results/final2_coefficients.csv`. All features are 0/1 indicators on one scale, so

we report β in minutes; standardising would only rescale each term by its frequency. On weekdays the 10–11h peak adds 5.5–5.9 min, precisely estimated. Saturday 12–16h adds 12–15 min, and the first day of a long weekend adds 15–21 min at 12–18h, the largest effects in the model; their intervals are wide because the training years hold only about ten first days, but all exclude zero. The last day of a long weekend is *faster* at 10–12h (−3.9 min), consistent with southbound demand having already left. The eve of a long weekend adds 8.3 min on Friday 20–22h and up to 14.6 min on other weekdays. Make-up workdays add 11–21 min at 10–16h (6 training days), like a busy Saturday. Statistical $\neq$ practical significance: the 01h term (−0.11 min) is significant but irrelevant to a driver. VIFs ≤ 3.1 : no collinearity concern.

Hypothesis tests (training years; OLS on log minutes, day-clustered SE). **H1 rejected, informatively**: first-day mornings are 18.4% *faster* than Saturday mornings, and the first-day excess appears only in the afternoon (+11.2%, $p=0.089$). Holiday traffic leaves later, so a single holiday flag would not do. **H2 mostly supported**: last-day effects are −0.3%, −0.6%, +0.4% in the morning, afternoon and evening, but 10–14h is 9.9% faster ($p = 0.0001$). **H3 supported**: eve evenings are +11.9% (Friday, $p < 0.001$) and +28.9% (other weekdays, $p = 0.0007$). **H4 suggestive**: −18.8% ($p = 0.028$ one-sided) in the log model, but the final-model interval covers zero.

Error analysis. The three largest 2025 errors (Table 8) are not holiday days: all are ordinary days under-predicted by about 44 min, with both the first segment and the tunnel section slow. Without incident, weather or road-work data the cause is unverified (incidents, closures or rain are plausible). The shared pattern is information the model never sees, so the first fix is better inputs (weather forecasts, announced traffic-control notices), not a

Feature set	CV MAE	Test all	Ordinary	LW 08–18h	First day 10–18h	Eve 16–23h
A: calendar only	1.56±0.33	1.68	1.54	5.34	14.46	6.36
B: A + top-30 holiday columns	1.55±0.26	1.55	1.47	3.41	6.30	5.59
C: full	1.41±0.23	1.42	1.35	2.96	6.25	7.83
C – day-type×block	1.56±0.33	1.64	1.52	4.79	13.38	5.50
C – Friday terms	1.45±0.23	1.46	1.39	2.96	6.25	7.83
C – eve terms	1.41±0.23	1.41	1.35	2.96	6.25	6.59
C – long-weekend terms	1.44±0.27	1.52	1.35	5.49	15.14	7.83
C – long-holiday (≥4 d) term	1.41±0.23	1.42	1.35	3.00	6.44	7.83
C – Saturday terms	1.51±0.35	1.56	1.51	2.98	6.27	7.83

Table 7: Feature ablation, MAE in minutes (same L1 solver and weights). Slice columns are hourly MAE within the slice.

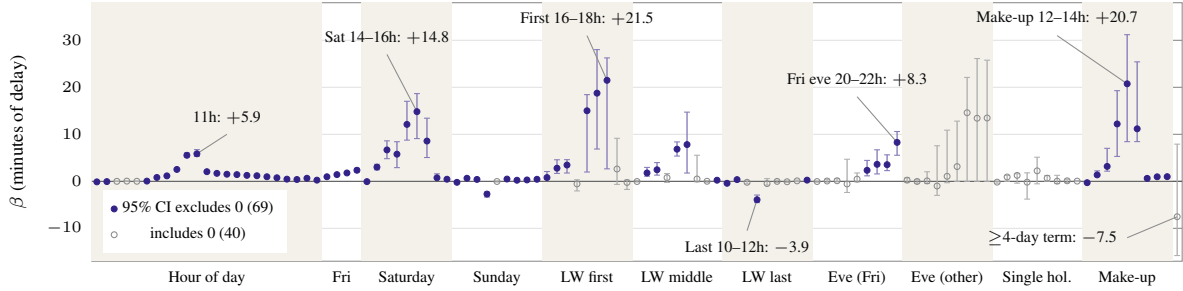


Figure 4: All 109 coefficients of the final model with 95% intervals (300 day-resampled refits). Within each day type (alternately shaded) the points run from the main effect through the eight two-hour blocks 06–22h. Every term is a 0/1 indicator, so β is in minutes and directly comparable across terms; all VIFs ≤ 3.1 .

#	Time (2025, ordinary day)	Actual	Pred.	Error	Last wk
1	07-01 Tue 10h	79.1	34.7	-44.4	35.0
2	08-16 Sat 11h	84.9	40.7	-44.2	65.7
3	09-12 Fri 11h	79.0	35.0	-44.0	32.4

Table 8: Largest 2025 errors (minutes; error = actual – predicted). In all three, km 0–5.5 takes 20–22 min and the tunnel section 48–54 min, while the whole trip takes about 30 min at free flow.

bigger model; for the calendar part, eve evenings (about ten training eves) are next.

9 Conclusion and Limitations

Answer to the research question: yes. Calendar and holiday structure alone let a 109-term median linear model forecast day-ahead hourly travel time with MAE 1.49 min on a pre-registered blind 2026 test (1.42 on 2025), 26% below last week’s same hour and 39% below it on long weekends; its recommended departure hour is within 0.45 min of the best on long weekends, inside the 1.0 min threshold. It ties gradient boosting overall and beats it where congestion occurs, while every term remains a readable number of minutes. What mattered is *when* in the day each holiday type is slow: first-day and Saturday afternoons, eve evenings. *Recommendation:* a trip app can use the forecast directly to

suggest departure hours before long weekends; for the first day, shifting a departure out of the 12–18h block saves the most time. *Limitations:* few holiday days (5 first days and 5 eves in 2026, 8 each in 2025) give wide intervals; eve evenings remain the weakest slice; the target ends 2.6 km past the Toucheng exit and sums segment times; 2023 ordinary days were down-weighted; no weather or incident data; the 1.0 min threshold is assumed. Next: weather and traffic-control inputs, eve terms extending to midnight, and the rest of 2026.

Contributions of Each Member

- 李尚哲 (314580057): §1, coordination.
- 呂丞頤 (314580055): references, §2.
- 黃駿甯 (314580075): data pipeline, §3.
- 吳美真 (414551035): EDA, H1–H4, §4.
- 孫傳康 (112550176): features, tuning, §5.
- 傅維雨 (112550149): results, blind test, §6–7.
- 張凱華 (315554024): coefficients, errors, §8.
- 李玠廷 (315554034): §9; Q&A preparation.
- All: slides, proofreading.

AI usage. Claude (Anthropic) wrote the scripts, ran the experiments and drafted the text. The team chose the topic and user, reviewed protocol and results (catching test-set leakage early) and is responsible for all content. [Version; checks by member.]

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